Research Report

Satellite Networks and Their Vulnerabilities

Shymus



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Introduction

The proliferation of satellite constellations, particularly in low Earth orbit (LEO), has profoundly reshaped the space landscape. These constellations deliver wide coverage, a significant reduction in latency and an overall improvement in communication performance. At the same time, the growing integration of satellites into terrestrial networks — notably within 5G architectures — strengthens service continuity and opens the way to new uses, from the Internet of Things to critical communications. This increasing convergence, however, comes with more complex architectures and a larger potential attack surface.

In this context, cybersecurity concerns take centre stage. Satellite communications rely on a combination of space and ground elements: ground stations, modems, routers, standard protocols and encryption mechanisms. While these components ensure performance and interoperability, they also introduce exploitable vulnerabilities — especially in an environment where the radio-frequency medium is open by nature. Securing satellite communications thus becomes imperative, both to protect data and to preserve the resilience of critical infrastructure.

Beyond purely technical aspects, this report adopts an approach oriented towards cybersecurity and Space Situational Awareness (SSA). It explores the mechanisms used to secure communications, from classical encryption standards to the perspectives offered by quantum cryptography, while highlighting the limits and challenges associated with each approach. This analysis follows on from earlier research and aims to deepen the understanding of satellite systems as technical, cyber and strategic objects.

CHAPTER 1

Introduction to satellite communications

1.1 Electromagnetic waves

Electromagnetic waves are the invisible link between Earth and space. They are not mere physical abstractions: they are the paths along which our messages, images and data travel to the satellites orbiting above us. Made of an electric field and a magnetic field that oscillate continuously, each perpendicular to the other, they advance together in a single direction at the speed of light. These oscillations cross the vacuum of space to carry a signal from a ground station to a satellite thousands of kilometres away.

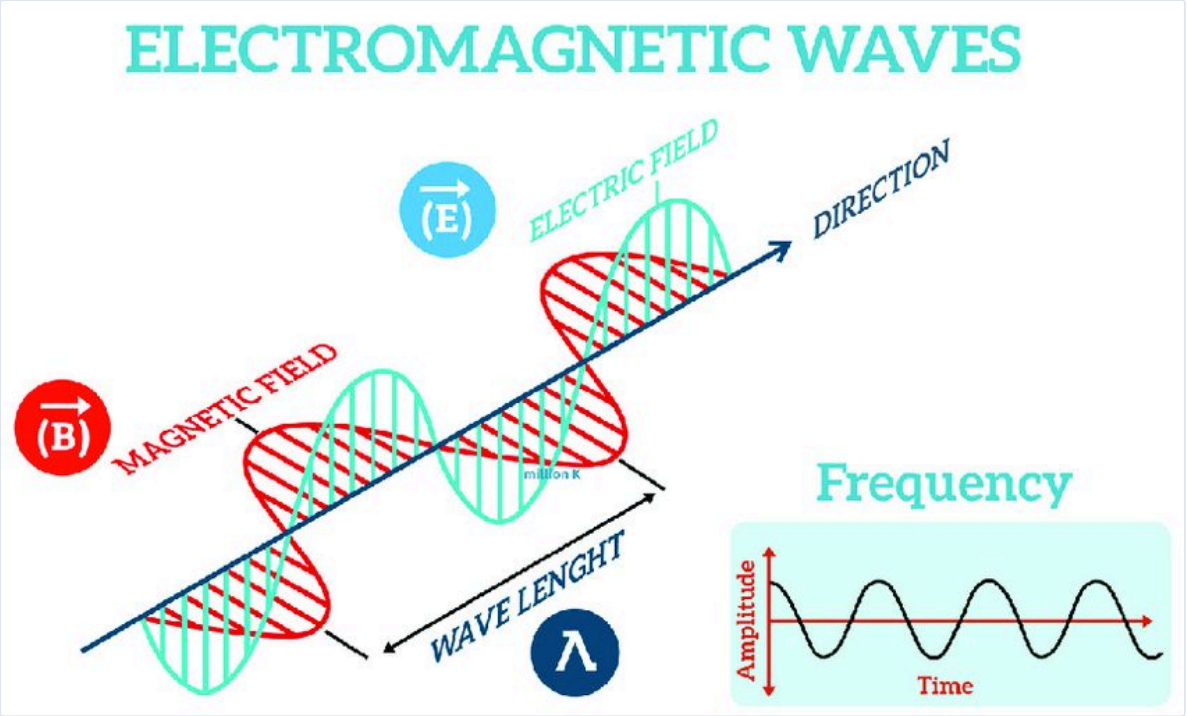


Figure 1.1 — Electromagnetic waves: coupled electric and magnetic fields propagating together.

Their behaviour rests on two fundamental parameters: frequency and wavelength. Frequency is the rhythm — the number of cycles per second; wavelength is the distance between two of those cycles. The two are inseparable, bound by the equation $c = f \times \lambda$, where c is the speed of light. The higher the frequency, the shorter the wavelength, and this simple relationship governs the entire organisation of the electromagnetic spectrum.

That spectrum is immense. It spans from radio waves, used for broadcasting and communication, to gamma rays, so energetic that they are used in medical treatment. Within this range every portion has its role: the microwaves that guide radar and carry satellite internet, the infrared that transports the Sun's heat, visible light that gives meaning to our perception, the ultraviolet, the X-rays. An entire invisible universe surrounds us. But for those passionate about space and satellites, it is above all the radio and microwave bands that

matter. They are what deliver GPS data to our phones, relay images of Earth from orbit, and carry the commands sent from a control centre to a satellite in the silence of space.

The frequency bands used by satellites are UHF, S, C, X, Ku and Ka. They act like invisible highways along which our communications travel. Each band has its own characteristics: some penetrate clouds better, others allow more data to be transmitted.

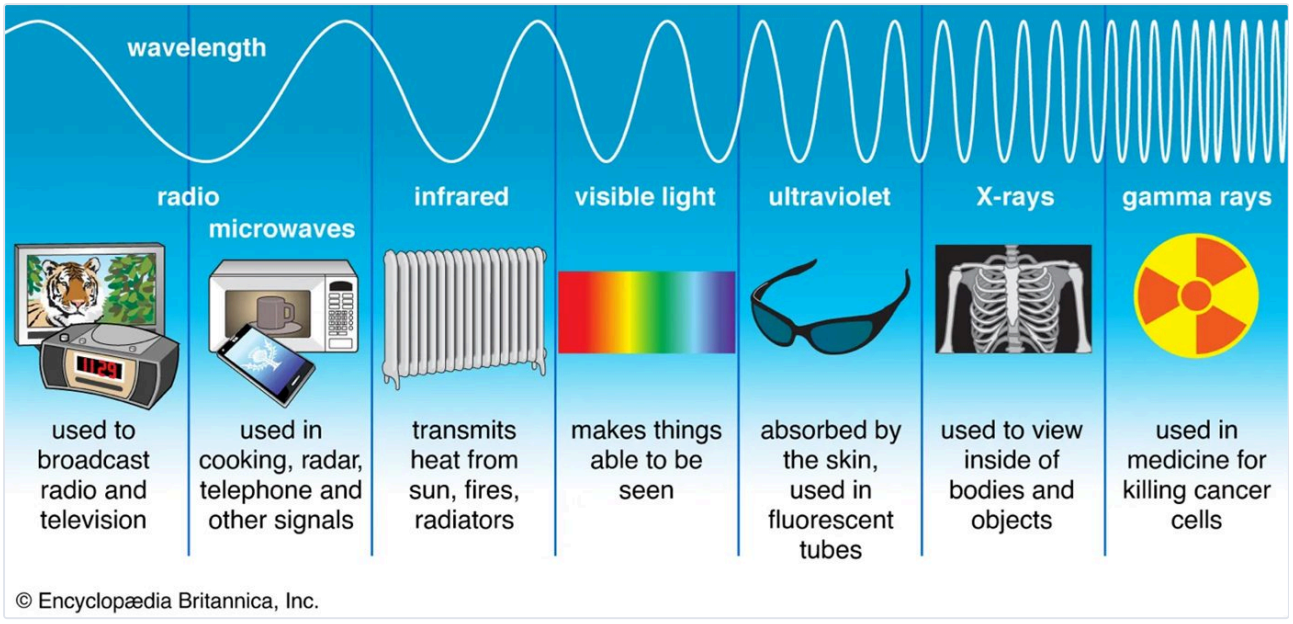


Figure 1.2 — The electromagnetic spectrum, from radio waves to gamma rays.

Ultimately, electromagnetic waves are more than a concept of physics: they are the key to modern space exploration. They embody the powerful link between Earth and its satellites, and remind us that, through these invisible oscillations, an entire dialogue with space becomes possible.

1.2 Satellite communication components

It all starts with the antennas. They capture and transmit radio waves — the ears and the voice of the satellite. They must maintain stable links, sometimes in extreme conditions, and often across different frequency bands (S, L, C, Ku, Ka).

Next comes the command and control system (C2). It handles the commands sent from the ground station and the telemetry data returned by the satellite. Every time a control centre adjusts attitude, triggers a manoeuvre or checks the health of the solar panels, it is the C2 that executes and reports.

Then come the transponders. Their role is to amplify and retransmit signals, sometimes shifting frequency to avoid interference. They give the satellite its function as an active mirror, sending back to Earth streams of information that have been transformed and reinforced, ready to be redistributed.

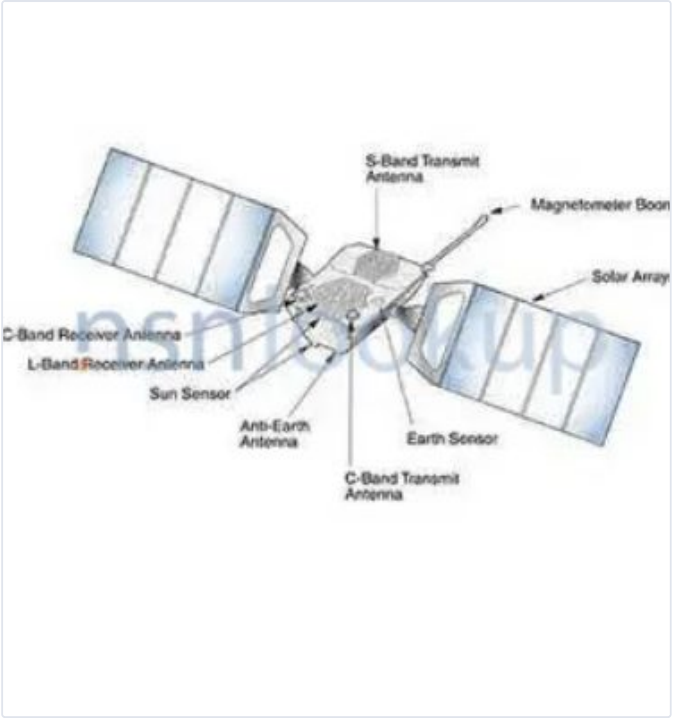


Figure 1.3 — Main components of a communication satellite.

The transmitter is responsible for converting data into carrier signals modulated on a precise frequency. It is the transmitter that sends images, GPS streams or secure communications. Conversely, the receiver acts as the fine ear, capturing RF signals, demodulating them, then separating the useful information from the noise and the carrier.

1.3 Types of satellite communication

When we talk about the types of satellite communication, we reach the very heart of the connection between Earth and space. Two major families dominate the field: radio-frequency communications and free-space optical (FSO) links.

The first, the most widespread and the most historic, is radio. These waves span a range from a few tens of MHz to several tens of GHz. They cross the atmosphere, bridge the distance between Earth and orbit, and return laden with information. This is the invisible channel through which our GPS signals, weather imagery and military and civil communications pass. What is striking is the robustness of these frequencies: they adapt to atmospheric conditions, propagate over long distances and allow continuous coverage. One could almost say they are the lifeblood of communication satellites.

But there is a second route: free-space optical communication (FSO). Here we no longer work in the classic radio bands but in optical frequencies, generally in the infrared around 1064 nm or 1550 nm. The advantage is enormous: a far higher data rate than radio waves. Gigantic volumes of information can be transmitted, almost at the speed of light, with surgical precision. But this gain comes at a price: the beam is much narrower, and therefore extremely sensitive to alignment and to atmospheric disturbances such as clouds or rain. It is a demanding technology, but one that opens the way to space communications of unprecedented capacity.

The diagram of atmospheric opacity is a reminder that the atmosphere acts as a natural filter. Some parts of the electromagnetic spectrum are completely blocked (X-rays or gamma rays, for example), while other windows — such as the "radio window" or the "optical window" — open to let waves through.

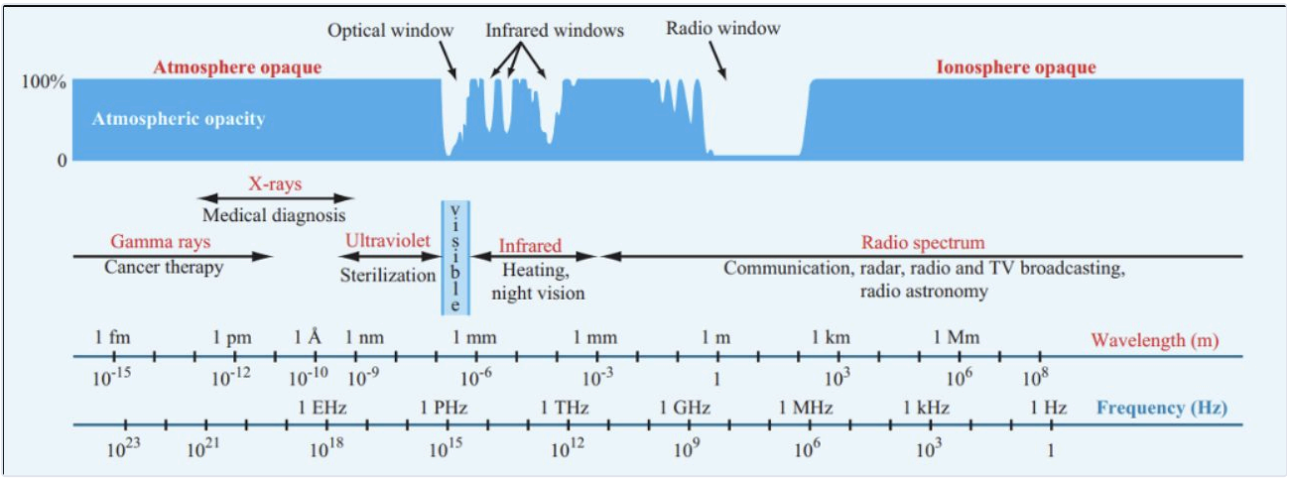


Figure 1.4 — Atmospheric opacity: the windows that let radio and optical signals through.

Let us see how these two approaches complement each other: radio frequencies for reliability and continuity, optical links for speed and data density. Two different voices: a deep, steady voice that carries far (radio waves), and a fine, precise voice that whispers data streams at very high speed (optical links).

1.4 Radio-frequency system

The radio-frequency communication system takes raw information — whether data, sound or images — and converts it into electromagnetic waves ready to travel through the vacuum of space. It is, quite literally, the voice of the satellite.

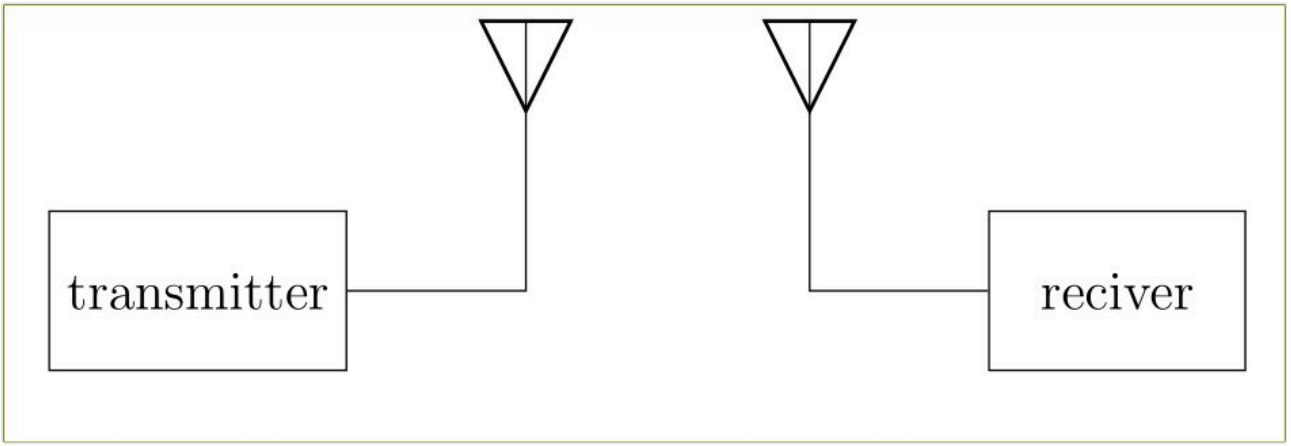


Figure 1.5 — The radio-frequency link: the satellite's free-space communication channel.

There is the free-space communication channel. This space, made up of the atmosphere and orbital vacuum, is not neutral: it can filter, attenuate or disturb signals. Yet, despite the obstacles, these waves manage to

travel thousands of kilometres to maintain the link with Earth.

Then comes the radio receiver. Its role is to capture the incoming signals, demodulate them, and turn what was only an electromagnetic vibration into readable data. Thanks to it, bits sent from space become weather images, GPS coordinates or internet streams.

Inside the transmitter, several stages follow one another like a well-oiled mechanism. The data interface is the entry point: this is where digital or analogue data enters the system. The modulator then encodes this data into a high-frequency wave, giving it a form suited to the journey. The power amplifier takes over to provide the necessary strength: the stronger the signal, the better it can cross space and the layers of the atmosphere. Finally, the antenna radiates this amplified signal, projecting it towards its destination with precision.

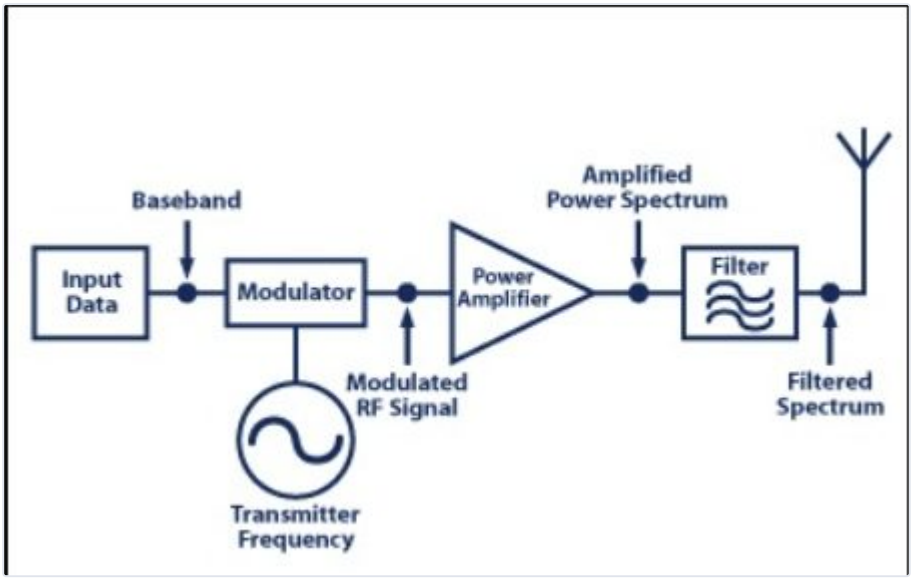


Figure 1.6 — Power amplification stage in the transmit chain.

The accompanying diagram makes this chain clear: raw data on the left, transformed into modulated signals, amplified, filtered, then sent out by an antenna. It is this mechanism that keeps our satellites in constant dialogue with us, and this invisible thread that holds together the whole of modern space adventure.

1.5 Radio-frequency system, part 2

When we dig a little deeper, everything begins on the receive side: the antenna captures oscillations that have travelled thousands of kilometres. Immediately afterwards, a Low-Noise Amplifier lifts this signal without adding unnecessary noise. This small component often makes the difference between a usable link and a lost signal. Then filters select the useful frequency window, amplifiers shape the power, and the demodulator "pulls" the information encoded in the carrier to reconstruct usable bits. It is a chain in which every link must be tuned to within a micro-decibel.

In terms of bands, one typically works between 30 MHz and 60 GHz, but this continuum hides fundamental choices. The low bands (VHF/UHF/L) offer better atmospheric penetration and wide coverage — perfect for robust, mobile or critical links — at the cost of limited throughput and a more crowded spectrum. The high

bands (C/X/Ku/Ka) offer very high data rates and more directive antennas (narrow beams, high gain): ideal for image downlinks or heavy traffic, but sensitive to rain and water vapour and demanding a precise aim (strict line-of-sight).

The atmosphere is an actor in its own right: it opens windows (radio, optical) and closes others. This is why we speak of atmospheric opacity. Engineers exploit these windows to guarantee availability and performance.

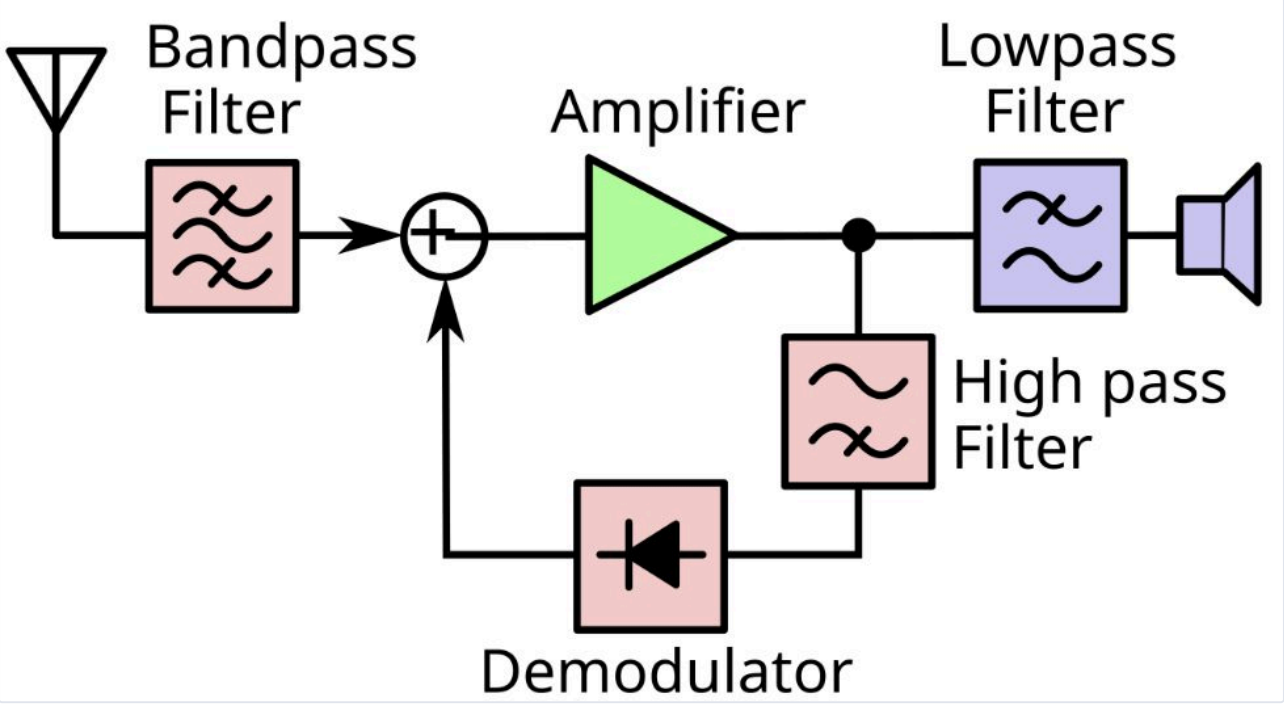


Figure 1.7 — Receive chain: antenna, low-noise amplifier, filtering and demodulation.

In practice, this translates into concrete decisions: installing a high-quality LNA (Low-Noise Amplifier) to improve the SNR (signal-to-noise ratio), choosing the Ka band for high-definition image transfer, keeping a redundant L-band link for critical commands, or using antenna systems that compensate for the dynamic pointing of a LEO satellite.

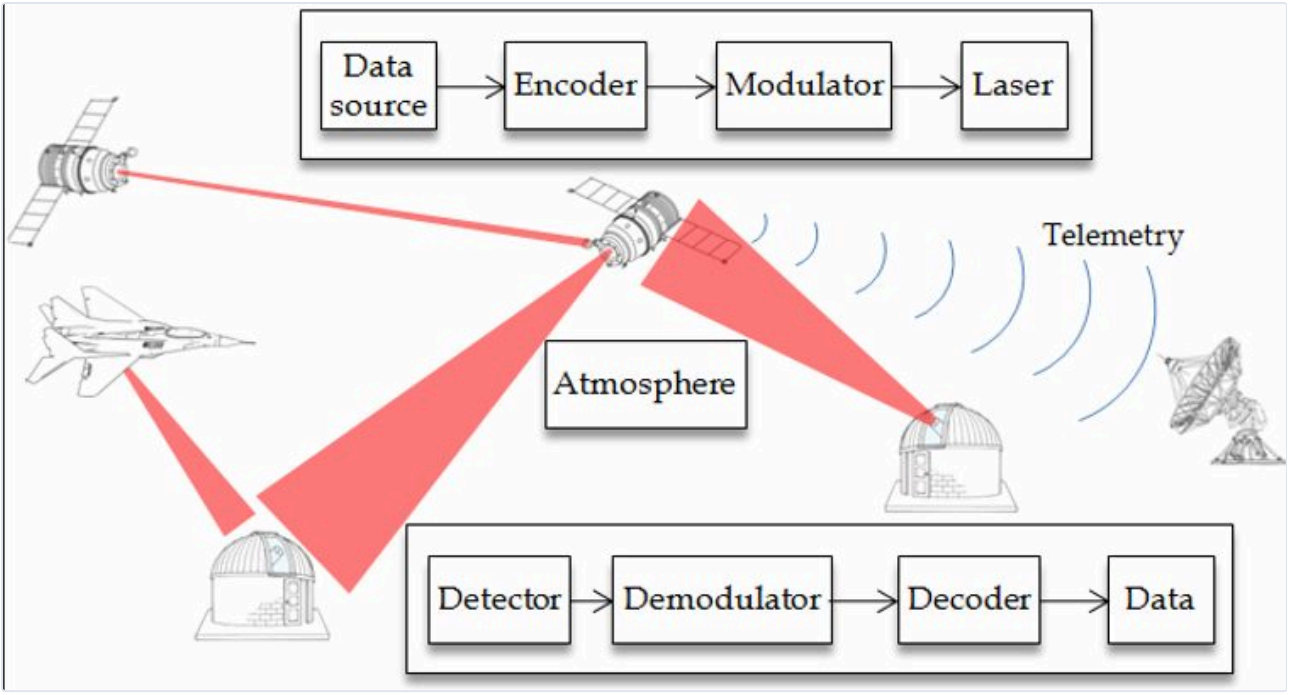


Figure 1.8 — A complete RF chain, from raw data to the radiated signal.

Designing an RF chain means seeking the balance between a signal that must survive disturbances and the desire to increase transmission capacity. Every optimisation — better filtering, suitable polarisation, clean amplification, a sharpened antenna design — translates into operational confidence: confidence that the telemetry will tell us the truth, that commands will arrive on time, that the image will be sharp.

1.6 What is a VSAT?

VSAT stands for Very Small Aperture Terminal.

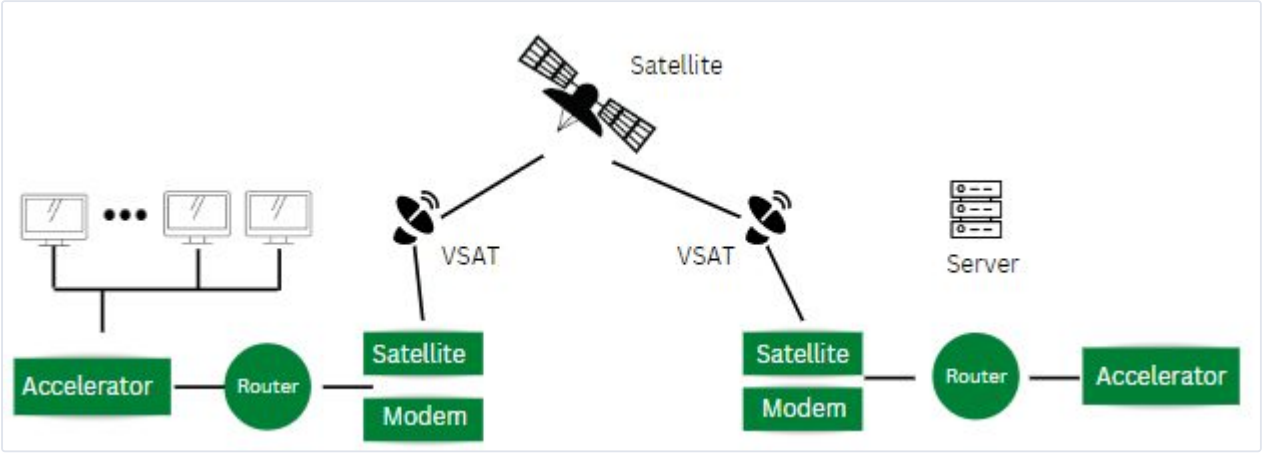


Figure 1.9 — A Very Small Aperture Terminal (VSAT).

A VSAT is used in small ground stations to transmit and receive data, telemetry and video over a satellite communication network. It includes a transceiver and an interface that allows connection to local networks.

A communication between two VSAT stations via a satellite illustrates that:

- User terminals are connected to a router.
- The router is linked to a satellite modem, sometimes accompanied by an accelerator to improve performance.
- Data is transmitted to the satellite, then relayed to another VSAT station.
- At the far end, data passes through a router and an accelerator before reaching the server.

1.7 Network architecture

VSAT networks have different types of architecture depending on how the terminals are organised.

VSAT star network: all VSAT terminals communicate through a central hub. The hub acts as the point of control and traffic redistribution. Communications between two VSATs necessarily go through the hub.

Advantages: centralised management; ease of supervision and control.
Drawbacks: dependence on the hub (single point of failure); higher latency for VSAT-to-VSAT communications.

VSAT mesh network: VSAT terminals communicate directly with one another. A central hub is not necessary. Each link operates on a dedicated channel.

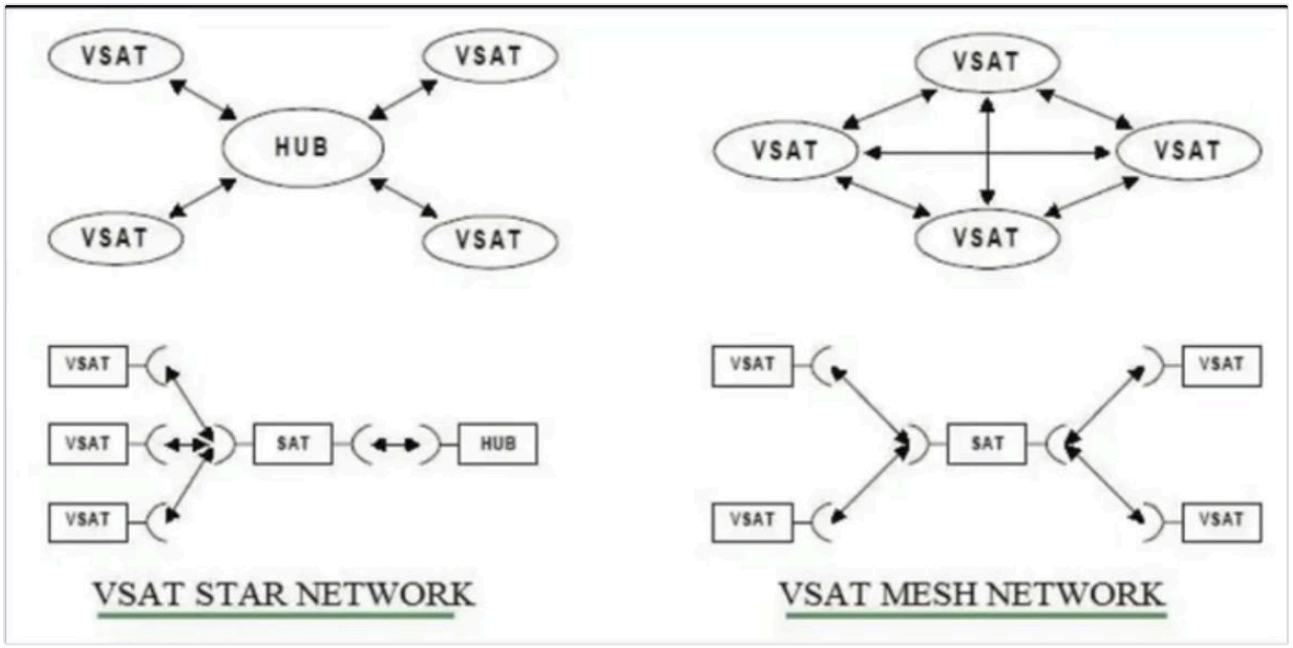


Figure 1.10 — VSAT network architectures.

Advantages: reduced latency; direct and more efficient communication.
Drawbacks: higher cost; more complex management.

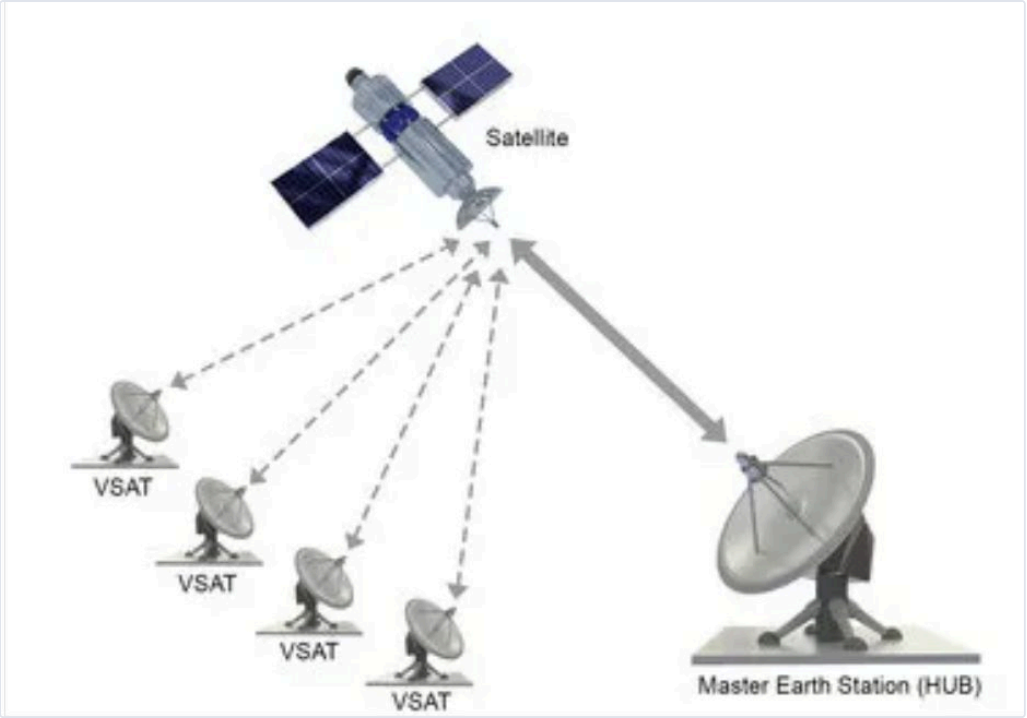


Figure 1.11 — VSAT star network built around a central hub.

This diagram illustrates a star VSAT architecture, in which a main ground station (HUB) provides centralised management of communications via a satellite. The satellite relays between the hub and all the remote VSAT terminals, enabling data transmission. In this configuration, the uplinks and downlinks are clearly identified between the hub and the satellite, underlining the central role of the main station in network control.

This star topology is widely used in VSAT networks because of its simplicity of deployment and its ability to optimise bandwidth use through centralised management. It also offers better overall visibility of the network and makes it easier to apply control and security policies. However, this architecture has certain limitations, notably a strong dependence on the hub — a single point of failure — and increased latency for inter-VSAT communications.

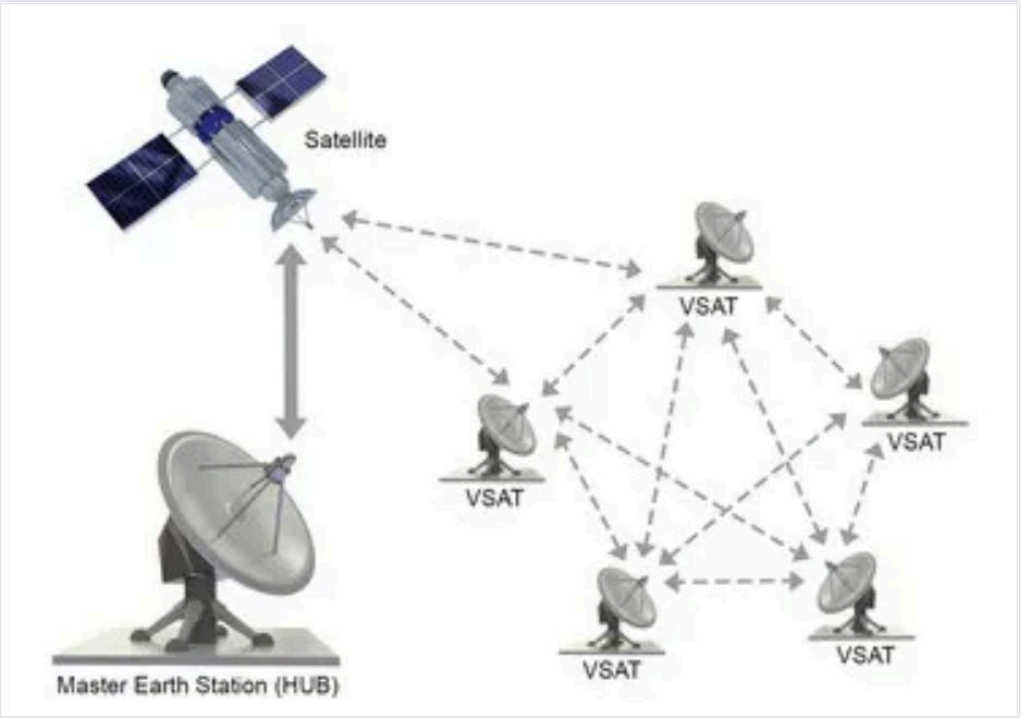


Figure 1.12 — VSAT mesh network with direct terminal-to-terminal links.

Here we have a VSAT communication architecture combining centralised management around a main ground station (HUB) with distributed communications between VSAT terminals, following a partially meshed topology. The satellite, positioned as the central element of the system, relays signals between the hub and the various remote terminals, playing a key role in distributing and collecting data flows.

The hybrid architecture combines the advantages of the star and mesh topologies. Centralisation around the hub ensures efficient supervision and simplified network administration, while the meshed links reinforce flexibility and operational resilience. This added complexity, however, also introduces security issues, linked to the multiplication of radio-frequency links and to the critical role of the hub and the satellite in the communication chain.

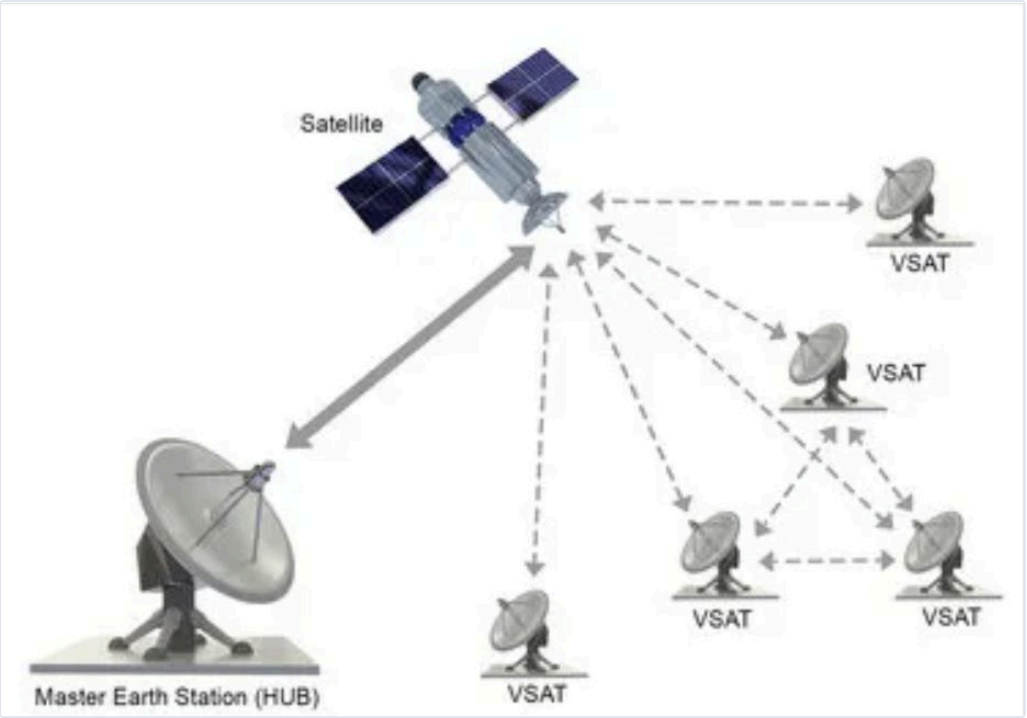


Figure 1.13 — Hybrid VSAT network combining star and mesh topologies.

Above we have a hybrid VSAT architecture that simultaneously integrates a star topology and a mesh topology, combining the advantages of both satellite communication models. In this configuration, a main ground station (HUB) provides centralised network management and communicates with the satellite over a bidirectional link. The satellite plays a fundamental relay role by broadcasting signals between the hub and the remote VSAT terminals.

From a security standpoint, this diagram highlights the need to integrate suitable protection mechanisms to face threats specific to hybrid VSAT networks, such as the eavesdropping of satellite communications, attacks targeting central nodes, and vulnerabilities linked to direct exchanges between remote terminals.

CHAPTER 2

Consumer satellite networks

2.1 Satellite infrastructure

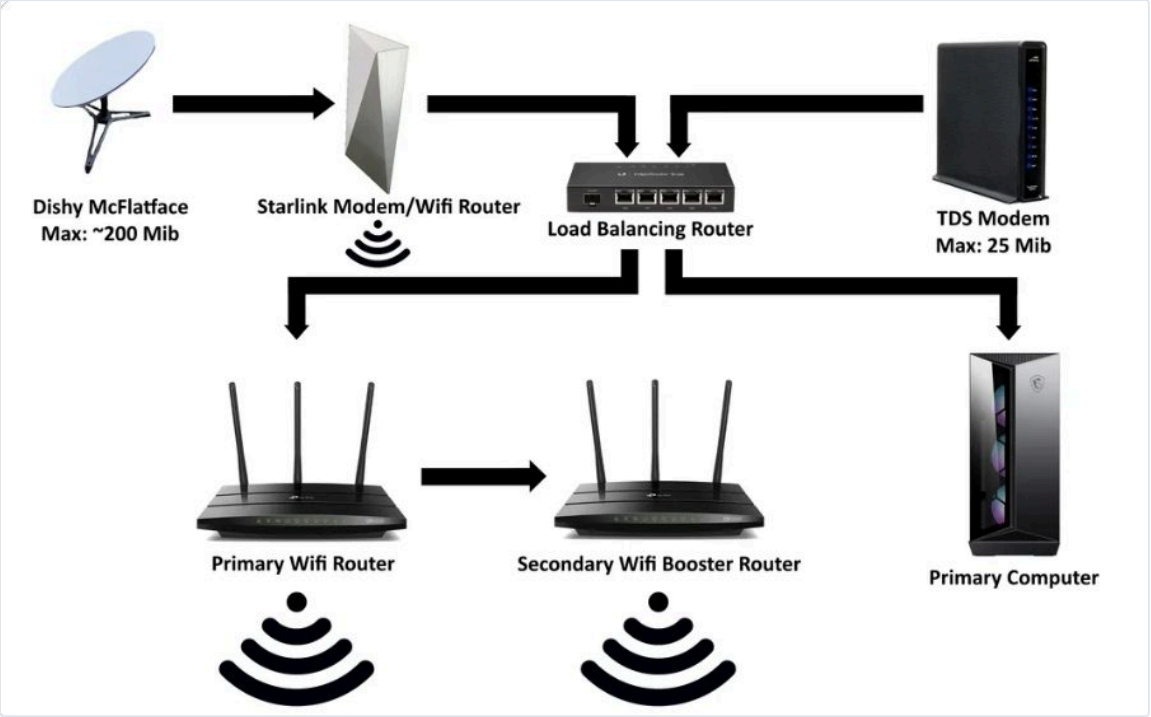


Figure 2.1 — A consumer-grade satellite network infrastructure.

Here is a consumer-grade infrastructure. The communication chain begins at the user terminal, represented by a receive-and-transmit satellite antenna, which establishes a bidirectional radio-frequency link with the satellite. This link is a channel that is open by nature, whose security relies mainly on encryption and authentication mechanisms implemented at the satellite modem. The modem plays a central role, providing both access to the satellite link and the distribution of traffic within the local network through routing and Wi-Fi functions.

Data flows are then directed towards intermediate equipment, such as a router, making it possible to manage several network links and to optimise service availability. From a cybersecurity point of view, these components are critical points, because they concentrate traffic and rely on software configurations that can introduce exploitable flaws in the event of poor management or insufficient updates.

The connection with other network equipment — notably primary and secondary Wi-Fi routers that provide connectivity for end terminals and connected objects — follows. This hierarchical architecture, from the satellite down to the user's devices, considerably widens the system's attack surface. Each intermediate device becomes a potential point of compromise, which could be exploited to intercept, alter or disrupt communications.

2.2 Satellite router



Figure 2.2 — A satellite router.

Satellite routers handle the flow of data over the satellite link, which is often characterised by high latency and limited bandwidth. Encryption of communications is presented as a key mechanism for protection against the interception of radio-frequency signals. However, the effectiveness of this encryption depends directly on the algorithms used, key management and the robustness of the software implementations, which may present exploitable vulnerabilities in the event of missing updates or weak configurations.

Satellite authentication and authorisation mechanisms aim to control access to network services and to prevent illegitimate connections. Nevertheless, these functions frequently rely on exposed administration interfaces, sometimes accessible remotely, which can be targeted by attacks such as authentication bypass, exploitation of default passwords or known software flaws. In this context, the router becomes a critical point whose compromise can grant extensive control over all satellite communications.

The presence of firewall and intrusion-detection functions is intended to filter traffic and identify abnormal behaviour. While these mechanisms strengthen network security, they remain dependent on suitable filtering rules and regularly updated signatures. In practice, limitations tied to the hardware resources of satellite routers, or to the specific nature of satellite traffic, can reduce the effectiveness of these devices and let targeted attacks or encrypted malicious flows slip through.



Figure 2.3 — A satellite modem.

Satellite modems carry out the modulation and demodulation of the signals passing over the satellite links. These processes, physical and software in nature, can introduce vulnerabilities when the implementations rely on proprietary firmware that is seldom audited or rarely updated. A compromise at this level can allow an attacker to disrupt communications, degrade the signal or, in some cases, inject malicious data into the transmitted flow.

From a communications standpoint, the modem is responsible for managing transmission parameters, such as frequencies, modulation schemes and synchronisation mechanisms with the satellite. A misconfiguration or malicious exploitation of these parameters can lead to targeted denials of service or facilitate passive interception attacks, particularly in scenarios where the encryption mechanisms are weak or poorly applied. Unauthorised access to the modem's management interfaces, whether local or remote, can give an attacker broad visibility over the communication flows and allow fine-grained manipulation of the transmissions.

2.3 Civil applications



Figure 2.4 — A satellite telephone.

An illustration of the diversity of civil applications that rely on space communication infrastructure. From a cybersecurity point of view, it highlights the gradual extension of these technologies into everyday uses, which leads to increased exposure to digital threats traditionally associated with terrestrial networks.

Satellite telephony systems, such as those used for voice and data communications on a global scale, rely on user terminals capable of establishing direct links with satellite constellations. Although these systems are designed to operate in isolated environments, their security depends heavily on the encryption and authentication mechanisms implemented at the terminal and the control network, which may present exploitable weaknesses in the event of missing updates or insufficiently robust design.

Satellite navigation systems (GNSS) are another pillar of consumer applications. Their use in civilian devices makes them particularly vulnerable to attacks such as jamming or signal spoofing, which can degrade the reliability of positioning data. These threats illustrate the limits of the security inherent in open GNSS signals, originally designed without advanced protection mechanisms.

Remote-sensing and satellite-imaging applications, although only indirectly accessible to the general public, also rely on data-processing and distribution chains that can be targeted.

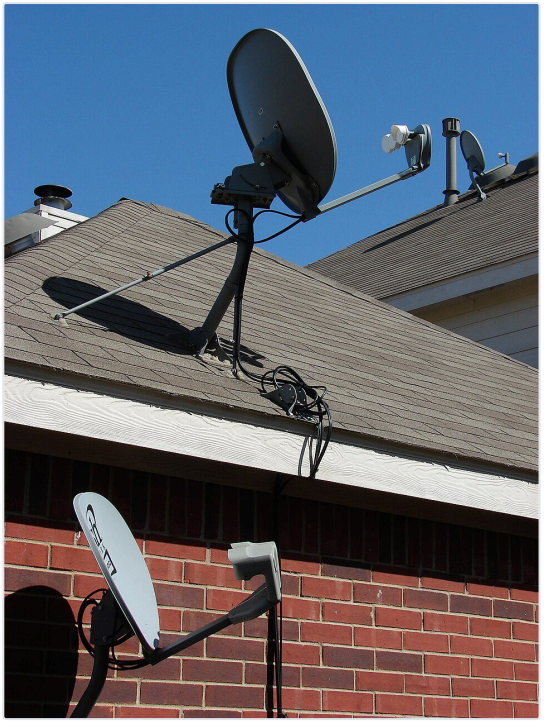


Figure 2.5 — A home satellite-television antenna.

Finally, direct-to-home (DTH) satellite television services rely on one-way broadcast architectures which, although less exposed to interactive attacks, can nonetheless be affected by interception techniques, content piracy or signal disruption. Taken as a whole, these consumer satellite systems illustrate the need to integrate cybersecurity considerations from the design stage, in order to face a growing attack surface tied to the democratisation of space technologies.

2.4 Satellite component user



Figure 2.6 — Affordable hardware for receiving satellite signals.

The components needed to listen to and receive satellite signals in a consumer context. From a cybersecurity point of view, this shows that access to satellite communications does not necessarily require complex industrial infrastructure, but can be achieved with affordable equipment — which considerably broadens the range of actors capable of interacting with these signals.

The antenna is the first element of this chain and plays a decisive role in capturing the radio-frequency signals emitted by a specific satellite. Any satellite transmission that is not properly protected can thus be received by third parties who have a suitable antenna.

The Software-Defined Radio (SDR) receiver — with its ability to tune to different frequencies and to perform demodulation in software — offers great flexibility, but also introduces risks. The use of an SDR allows a potential attacker to analyse, record and process satellite signals at low cost, facilitating passive-eavesdropping attacks or the reverse engineering of communication protocols.

As a whole, the security of satellite communications cannot be approached solely from the angle of controlling the space infrastructure. The availability of consumer tools for listening to and analysing signals calls for a security approach based on robust cryptographic mechanisms and secure protocol design, in order to limit the risks of interception and unauthorised exploitation.

2.5 How to listen in



Figure 2.7 — A reception antenna.

The first step is to identify the category of satellite to which the targeted signals belong:

- amateur-radio satellites,
- meteorological or scientific satellites.

These often broadcast publicly accessible data and are prime entry points for analysis and learning. Conversely, commercial and military satellites rely on more advanced protection mechanisms, but their presence within the same orbital infrastructure underlines that the level of security varies greatly depending on the use case and the threat model considered.

Selecting a specific satellite then relies on open databases that list active satellites, their orbits and their transmission parameters. Identifying the satellite's operating frequency is a critical step, because it allows the receiving equipment to be configured precisely. The frequency bands used by satellites are regulated and standardised, which makes their observation technically accessible. From a security standpoint, this underlines that confidentiality cannot rest on the secrecy of frequencies, since these are by nature observable by any suitable receiver.

Finally, taking into account the moment when the satellite passes over the listening area highlights the time constraint tied to low-orbit satellites. Trajectory-prediction tools, widely available, make it possible to anticipate these passes with precision. This accessibility reinforces the idea that eavesdropping on satellite communications rests on a combination of public information and consumer technology, rather than on means reserved for state actors.

CHAPTER 3

Constellations

3.1 Designing a constellation

The design of satellite constellations is today a central element of space communication and observation systems. Unlike traditional architectures based on a limited number of isolated satellites, constellations rely on a coordinated set of similar satellites, distributed according to carefully defined orbital parameters in order to meet a common objective. This approach ensures wide coverage, better service availability and, in some cases, a significant reduction in latency.

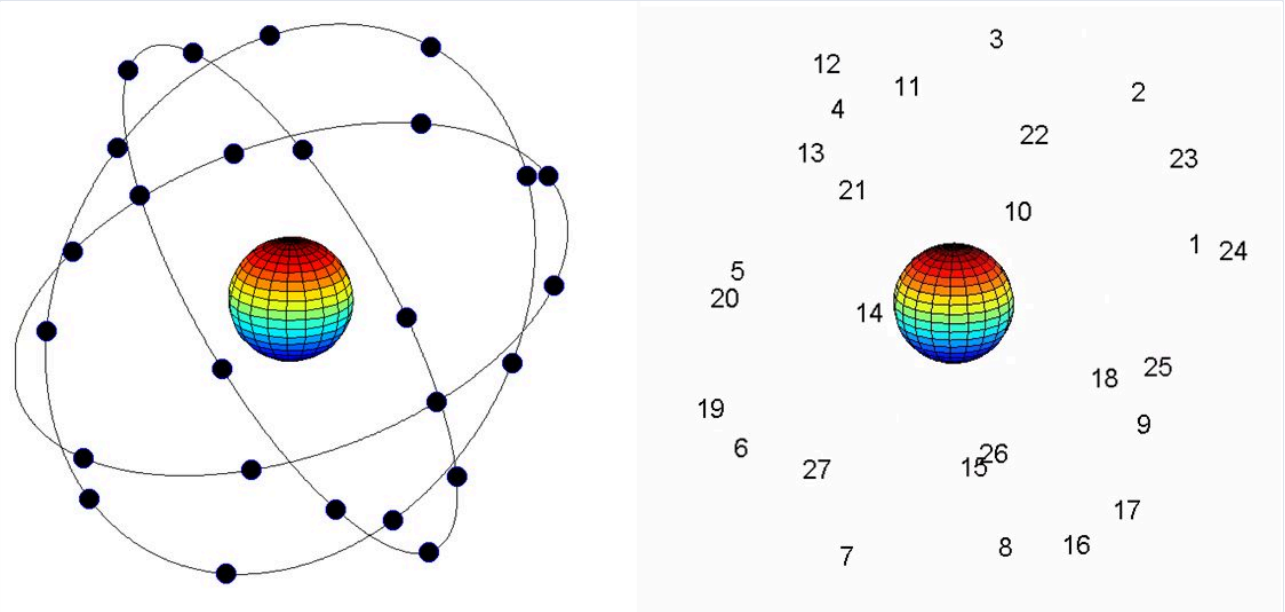


Figure 3.1 — Inter-satellite links within a constellation.

Modern constellations, particularly those in low Earth orbit (LEO), involve a large number of satellites operating in an increasingly congested space. The absence of strict universal standards for allocating objects in orbit complicates coordination between operators and increases the risk of collisions. This orbital density calls for rigorous planning of orbital planes and continuous monitoring of trajectories in order to preserve the safety and sustainability of space infrastructure.

Orbital perturbations must also be taken into account in constellation design. Satellites are subject to various disturbing forces, such as residual atmospheric drag, solar radiation pressure and non-uniform gravitational influences. These phenomena can cause gradual orbital drift, affecting the relative position of satellites within the constellation. Accounting for them is essential to maintain coherent coverage and to guarantee the quality of the services provided.

In the context of this report, this diagram highlights that the design of satellite constellations is not solely a matter of technical or operational considerations, but is also a decisive factor for the security and resilience of communication systems. The growing complexity of constellations, combined with orbital congestion,

increases the risk surface — both in terms of accidental disruptions and intentional threats. Constellation design must therefore integrate monitoring, coordination and mitigation mechanisms from the outset in order to ensure the durability and security of satellite services.

3.2 Constellation layout

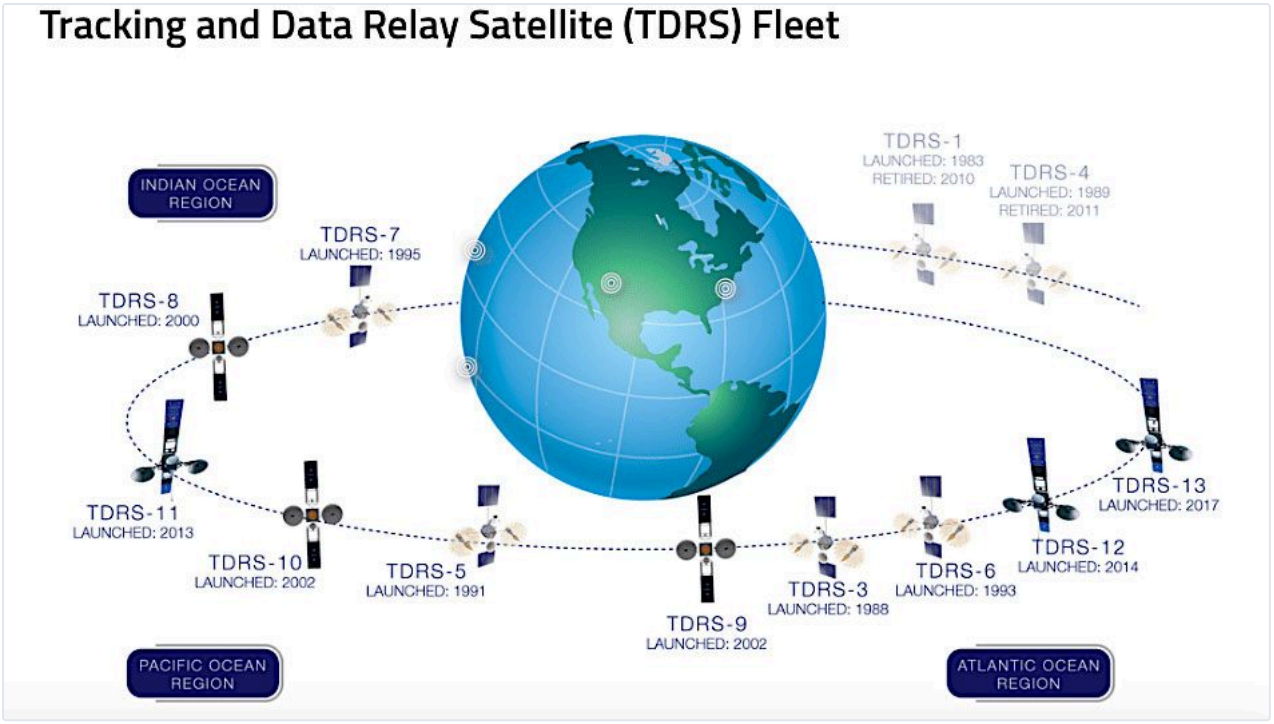


Figure 3.2 — Constellation design with relay satellites.

Relay satellites act as intermediaries between satellites in low or medium orbit and ground stations, by receiving, temporarily storing and retransmitting data to other nodes of the network. This architecture allows satellites that are not directly visible from a given ground station to maintain an active communication link, thereby improving the availability and reliability of space services.

An emblematic example of this type of system is the Tracking and Data Relay Satellite System (TDRSS), coordinated by NASA. This network relies on a constellation of geosynchronous satellites positioned so as to cover vast oceanic and continental regions. By maintaining a near-fixed position relative to the Earth's surface, these satellites ensure continuous coverage and allow the relay of data from space missions spread across different orbits.

3.3 Global coverage

Global-coverage systems generally rely on satellites in geostationary orbit, which offer permanent visibility over specific regions, or on constellations of satellites in low orbit, whose orbital motion provides dynamic, renewed coverage. In both cases, the aim is to guarantee service availability over vast geographical areas, including isolated regions or those lacking terrestrial infrastructure.

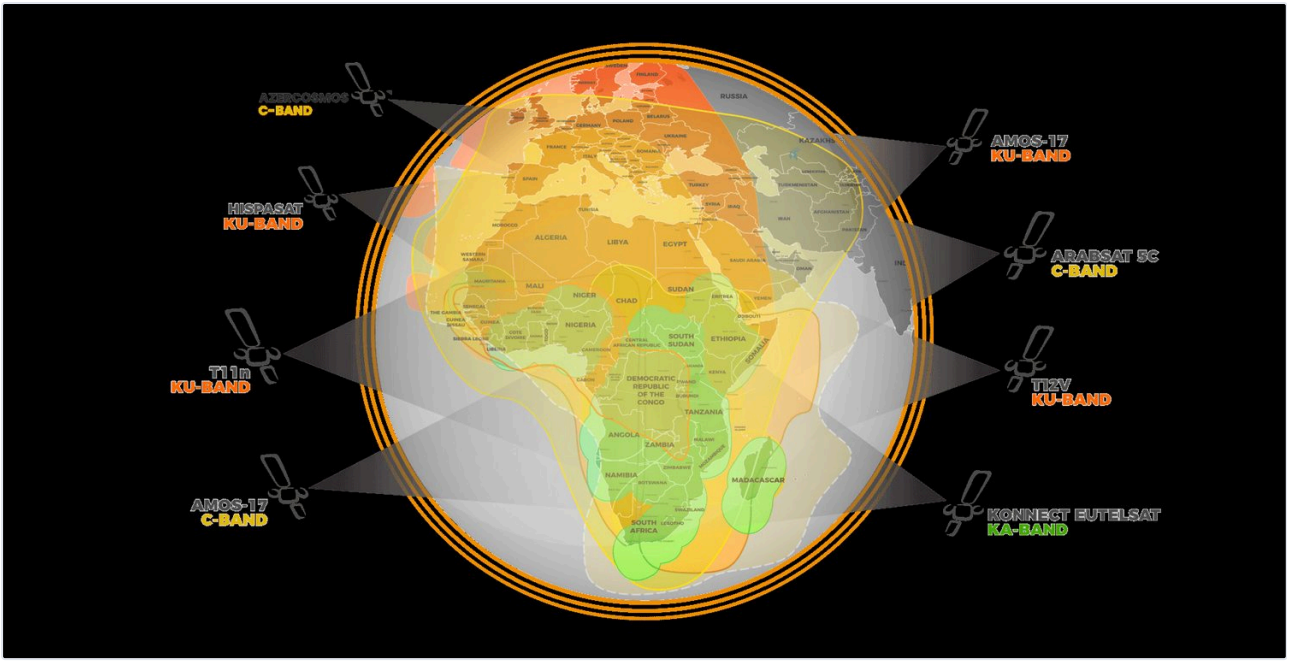


Figure 3.3 — Global connectivity provided by a satellite constellation.

Different frequency bands, such as Ku, Ka or C, are used to adapt communications to the needs of the services provided. The choice of these bands directly influences capacity, beam reach and resistance to atmospheric disturbances, making it a key parameter in the design of global-coverage systems.

However, providing coverage on a planetary scale involves technical trade-offs. Satellites at high altitudes, notably in geostationary orbit, cause greater latency and lower spatial resolution compared with low-orbit systems. These constraints must be taken into account when designing the applications and services that rely on this infrastructure. This global-coverage capability, although essential for many applications, also raises issues of performance, resilience and security in satellite communications.

3.4 Targeted-area coverage

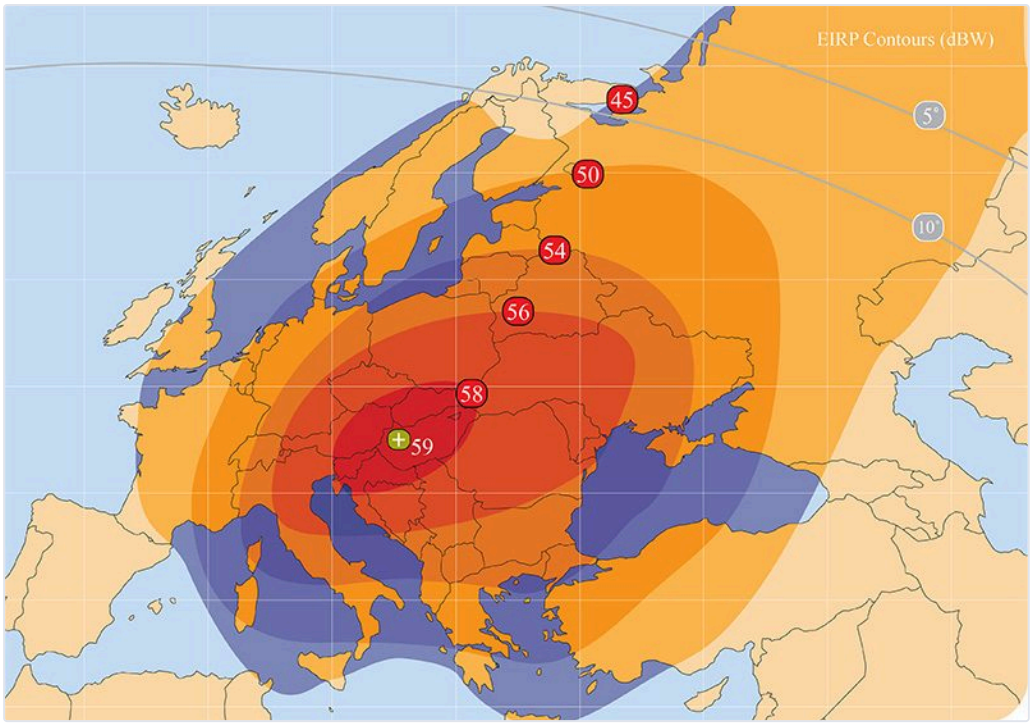


Figure 3.4 — Targeted coverage: the Amos-3 satellite.

Targeted coverage can be achieved through different orbital configurations, notably geostationary, low (LEO) or medium (MEO) Earth orbits. The choice of orbit directly influences the size of the covered area, the duration of the satellite's visibility and the spatial resolution of the data provided. By concentrating the beams on a given region, it becomes possible to optimise spectrum use and improve service quality for the users located in that area.

One of the main advantages of this approach is the ability to refine coverage in order to increase spatial resolution and the precision of observations or communications. This characteristic is essential for applications that require detailed monitoring, such as environmental surveillance, the management of critical infrastructure or certain security applications. In return, this specialisation implies a limitation of the service area and, in the case of non-geostationary satellites, shorter contact times with a given region.

While targeted coverage makes it possible to optimise performance and resolution over specific areas, it also introduces operational constraints that influence the overall design of satellite systems.

CHAPTER 4

Security of satellite communications

4.1 Future satellite communication methods

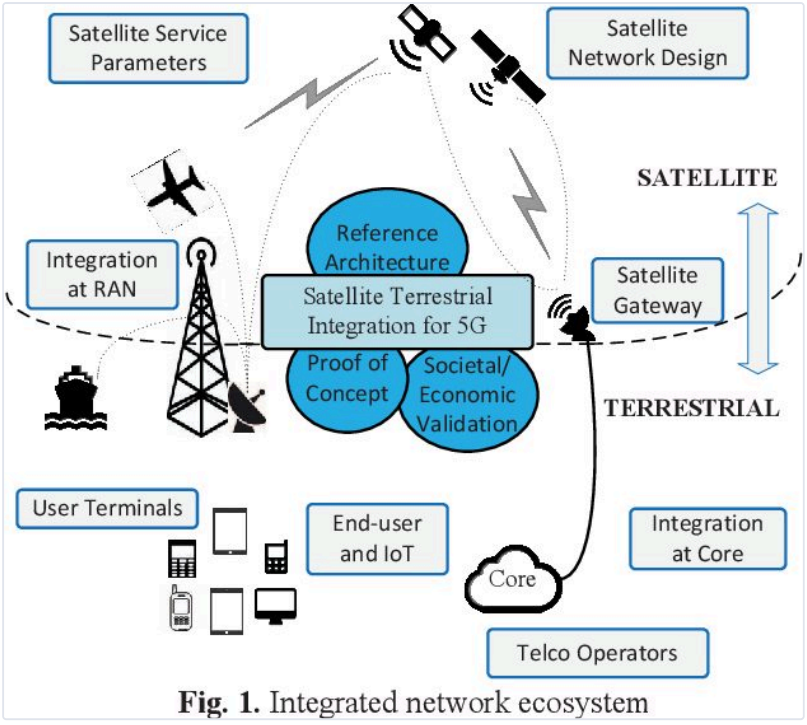


Figure 4.1 — Future satellite communication methods and the 5G reference architecture.

One of the major axes today is the integration of satellites within the 5G ecosystem. This approach aims to extend the coverage of terrestrial mobile networks thanks to the capabilities of satellite systems, particularly in isolated or poorly served areas. The notion of a reference architecture is central, in which the space and terrestrial segments interact closely, from user terminals and connected objects to the core of telecom operators' networks. This convergence increases the flexibility and reach of networks, but also introduces new dependencies between heterogeneous infrastructures.

Reducing latency is another central challenge, particularly for low-Earth-orbit (LEO) constellations. Recent developments aim to minimise transmission delays so as to make satellite services compatible with latency-sensitive applications, such as real-time communications or certain industrial uses. This evolution relies on denser constellations, optimised inter-satellite links and closer integration with terrestrial networks.

The growing importance of encryption standards in future satellite communications is clear. Faced with the rising volume of data exchanged and the diversification of uses, modern and robust cryptographic mechanisms become indispensable to guarantee the confidentiality, integrity and authenticity of communications. This standardisation strengthens the overall security of satellite systems while ensuring their interoperability.

Developments linked to quantum encryption, oriented towards the longer term, explore the use of quantum technologies to secure satellite communications. Although these solutions are still at an experimental stage, they represent a promising avenue for addressing future threats to classical cryptographic mechanisms, particularly in a context of rising quantum-computing power.

4.2 5G networks

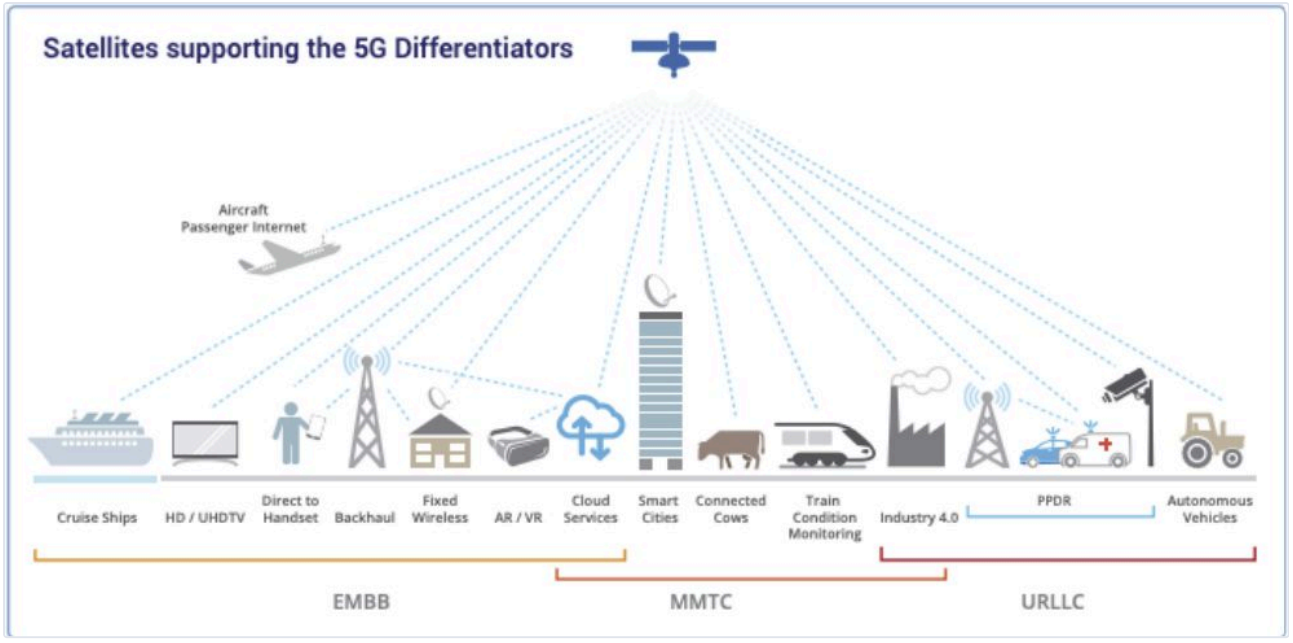


Figure 4.2 — Satellites supporting 5G use cases.

5G, as the fifth generation of mobile networks, offers high data rates, reduced latency and massive support for connected devices. Satellites can support these objectives by providing transmission links for a variety of uses, such as maritime and aeronautical connectivity, high-speed services in rural areas, or industrial and smart-city applications. This integration relies on a hybrid architecture in which the space and terrestrial segments cooperate closely.

The diversity of use cases supported by this integration covers high-speed services (eMBB), massive communications for the Internet of Things (mMTC) and critical applications requiring very low latency and high reliability (URLLC). The participation of satellites in these different scenarios underlines their growing role in the next-generation telecommunications ecosystem.

However, for satellites to fully meet the requirements of 5G — in particular for delay-sensitive applications — the question of latency becomes central, which explains the growing interest in low-Earth-orbit constellations.

4.3 LEO latency reduction



Figure 4.3 — Latency reduction with LEO constellations.

Compared with geostationary satellites, and because of their lower altitude and therefore shorter propagation distances, latency reduction is one of the major advantages of LEO constellations. This proximity, however, implies rapid orbital motion, requiring advanced mechanisms to maintain stable connections with ground stations and user terminals.

Digital signal processing and beamforming techniques are another key element. By concentrating radio energy on precise areas and limiting interference, these technologies improve link quality and optimise the use of spectral resources. They also allow better management of multiple simultaneous connections — an essential characteristic of dense low-orbit constellations.

As LEO constellations deliver faster and more responsive communications, the question of data protection becomes central, underlining the need to improve encryption standards.

4.4 Improving the encryption standard

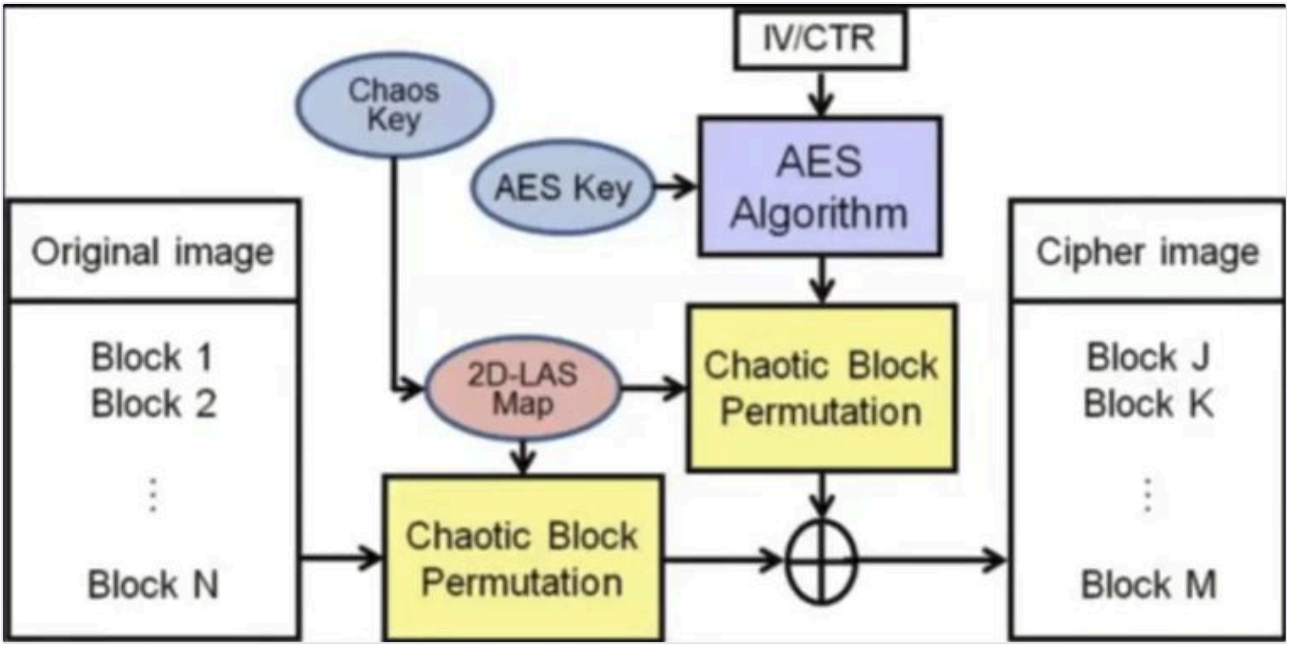


Figure 4.4 — An encryption architecture based on AES and chaotic permutation.

An encryption architecture based on recognised algorithms — in particular AES (Advanced Encryption Standard) — combined with chaotic permutation mechanisms. The use of encryption modes such as CTR (Counter Mode) makes it possible to secure data flows effectively while maintaining good performance, which is essential in the constrained environments of space systems. Chaotic permutation techniques add an extra layer of complexity, making attacks based on statistical analysis or correlation more difficult.

At the institutional level, bodies such as the CCSDS (Consultative Committee for Space Data Systems) play a central role in defining common security standards for space missions. These standards aim to guarantee interoperability between satellites, ground stations and operators, while ensuring a consistent level of security on an international scale. NASA, along with other space agencies, actively participates in these initiatives to strengthen the resilience of space communications against cyber threats.

While current encryption standards such as AES offer a high level of security for satellite communications today, the emergence of quantum computing raises new challenges that motivate the exploration of fundamentally different encryption mechanisms.

4.5 Quantum encryption

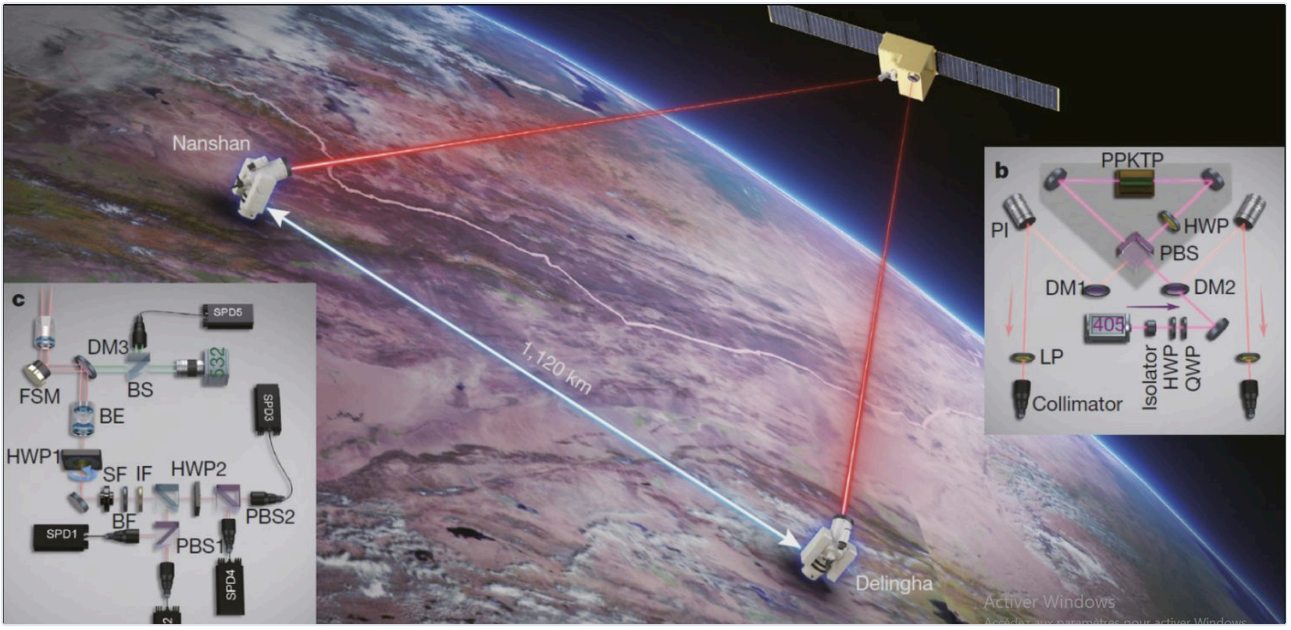


Figure 4.5 — Quantum key distribution: the Micius satellite.

In this context, satellites play a key role as relays for Quantum Key Distribution (QKD) over long distances, which is difficult to achieve through terrestrial infrastructure alone because of optical losses. Optical links between a satellite and several ground stations use polarised photons to transmit cryptographic keys whose integrity can be verified in real time. Any attempt at interception alters the quantum state of the photons, making an eavesdropping attack immediately detectable.

The projects mentioned illustrate the growing maturity of this technology. The Chinese satellite Micius, launched in 2016, demonstrated the operational feasibility of quantum-encrypted communications from orbit, marking a major advance in the field. For its part, the European Space Agency is developing the Eagle-1 project, the first European satellite dedicated to QKD, with the aim of integrating these security mechanisms into future pan-European communication infrastructure.

From a cybersecurity point of view, quantum encryption represents a paradigm shift: security no longer rests solely on the mathematical complexity of algorithms, but on intrinsic physical properties. This approach, however, still raises technical and operational challenges, notably in terms of cost, large-scale deployment, compatibility with existing networks and securing the ground segments, which remain critical points of the communication chain.

CHAPTER 5

Conclusion

5.1 Summary

The aim of this work was to analyse and structure the elements needed to design an architecture capable of discreetly capturing satellite signals and exploiting them in order to produce reliable leads for detecting and tracking low-visibility spacecraft. Through the study of satellite architectures, constellations, coverage mechanisms and the issues of performance and security, it becomes clear that this problem can only be addressed through an approach combining communications engineering, signal processing and cybersecurity.

The analysis of the various topologies and coverage modes, whether global or targeted, shows that satellite constellations — particularly in low orbit (LEO) — offer a favourable framework for the indirect observation of space objects. The growing density of these constellations, coupled with data-relay mechanisms and intelligent information distribution, makes it possible to increase the probability of capturing weak, intermittent or non-cooperative emissions. In this context, passive listening to satellite signals is a discreet approach, limiting the electromagnetic footprint of the observation architecture while exploiting already existing infrastructure.

The reliability of these leads also rests on mastering security mechanisms. The study of current encryption standards highlights a paradox: while encryption strengthens the confidentiality of communications, it is also an exploitable marker for cyber-electromagnetic analysis, notably through the study of metadata, traffic patterns and the physical characteristics of signals. In the longer term, the emergence of quantum encryption introduces new challenges — rendering certain forms of interception ineffective — but also new opportunities for analysis based on the very physics of quantum exchanges.

In conclusion, designing an architecture capable of discreetly capturing satellite signals and detecting low-visibility spacecraft rests on the convergence of several fields: constellation architectures, advanced reception and signal-processing techniques, latency management and the analysis of security mechanisms. This work shows that, in an increasingly dense and complex space environment, the cyber-electromagnetic approach is a relevant lever for improving Space Situational Awareness, while laying the foundations for future research oriented towards passive detection, multi-source data fusion and the behavioural analysis of space systems.

5.2 What comes next

Following on from this work, the next research step will turn towards a deeper analysis of satellite communications themselves, through approaches based on reverse engineering and signal decoding.